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$$\begin{aligned}2(3^x + 2^{x-1}) &= 3(2^x - 3^{x-1}) \\ \Rightarrow 2 \cdot 3^x + 2 \cdot 2^{x-1} &= 3 \cdot 2^x - 3 \cdot 3^{x-1} \\ \Rightarrow 2 \cdot 3^x + 2^x &= 3 \cdot 2^x - 3^x \\ \Rightarrow 2 \cdot 3^x + 3^x &= 3 \cdot 2^x - 2^x \\ \Rightarrow 3 \cdot 3^x &= 2 \cdot 2^x \\ \Rightarrow 3^{x+1} &= 2^{x+1} \\ \Rightarrow x + 1 &= 0 \\ \Rightarrow x &= -1\end{aligned}$$

References